The Optimality of Uniform Privacy Budget Allocation in Byzantine-Resilient Federated Learning for Anti-UAV Detection

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University
January 2026.

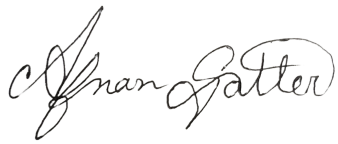
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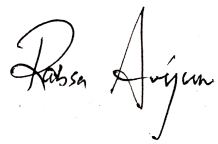
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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

Differentially private federated learning (DP-FL) requires each client to spend a finite privacy budget across communication rounds, and a growing literature proposes learning *how* to spend it, allocating more noise to rounds that tolerate it and less to rounds that do not. We show that for the dominant class of DP-FL mechanisms this optimisation is vacuous. Under per-sample gradient clipping, each round's contribution to the model is bounded by the clipping norm C irrespective of the noise scale, so the allocation influences only the noise term; minimising the accumulated noise variance subject to a fixed budget is then a convex problem whose solution is the *uniform* allocation. Any non-uniform schedule is strictly worse, and the penalty grows with the distance of the schedule from that minimum.

We validate this on federated thermal anti-UAV detection with a 566,789-parameter differentially private detector. Across 13 distinct allocations spanning three unrelated schedule families, power-law, bang-bang, and random draws from the budget simplex, a single scalar, the accumulated noise variance $\sum_t \sigma_t^2$, orders every outcome (Spearman $\rho = -0.767$, $p = 0.0022$). Uniform sits at the analytic minimum and is never beaten: the best alternative we found is bounded above by $+0.0003$ $\mathrm{IoU}_{.25}$ at n=16 paired trials with common random numbers. The same inequality closes allocation across layers and across clients.

We then study the security of this architecture under model poisoning and report a second consequence of high dimensionality. The camouflage on which DP-exploiting poisoning attacks depend, hiding adversarial perturbation inside the legitimate noise envelope, requires the honest update norm to be variable. In D dimensions a Gaussian norm concentrates as $1/\sqrt{2D}$, and we measure a relative standard deviation of 0.10% at D = 566,789, matching theory. DP noise therefore makes the update norm *more* predictable, capping the amplitude available to an attacker: the worst undefended degradation we observe at 30% malicious clients is 2.7% relative. Combined norm-plus-accuracy detection removes it entirely, at the cost of rejecting 28% of honest contributions, whereas classical Byzantine-robust aggregators are counterproductive under DP: coordinate-wise median costs -0.208 $\mathrm{IoU}_{.25}$ *with no attacker present*, because averaging is precisely the operation that suppresses DP noise.

Keywords: Federated Learning; Differential Privacy; Privacy Budget Allocation; Byzantine Resilience; Model Poisoning; UAV Detection

Dedication

Thanks to Almighty Allah for giving us the strength and patience to complete this thesis. We would like to dedicate this thesis to our beloved Supervisor Dr. Muhammad Iqbal Hossain sir, RA Fahmid UL Alam Juboraj for unwearied support, encouragement, and understanding throughout the academic journey. Their unwavering belief in us and our dedication and our believe in us has been a constant source of motivation.

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Chapter 1

Introduction

1.1 Background

Federated learning (FL) trains a shared model across decentralised clients without pooling raw data [6]. It does not, by itself, provide a privacy guarantee: model updates leak information about the data that produced them, and reconstruction, membership-inference and attribute-inference attacks all operate on shared gradients alone. Differential privacy (DP) supplies the missing guarantee by clipping each per-sample gradient to a norm C and adding calibrated Gaussian noise [2], at a cost in utility.

In the federated setting the privacy budget is consumed over many communication rounds, which raises a scheduling question: given a total budget ρ, how should it be divided among rounds? The intuition motivating adaptive schemes is that gradients are large early in training and small later, so early rounds should tolerate more noise. Several recent works formalise this as a learning problem and train reinforcement-learning agents to allocate the budget [11], [13].

This thesis asks whether that intuition survives contact with the mechanism it is built on, in a demanding application: federated thermal detection of small unmanned aerial vehicles, where targets occupy a handful of pixels and the detector has little accuracy to spare.

1.2 Motivation

Two observations motivate the work.

First, the adaptive-allocation intuition is stated in terms of *gradient magnitude*, but DP-SGD clips every per-sample gradient to a fixed norm before noise is added. If clipping binds, and we measure it binding by a factor of 27 in our own setting and by a factor of 8 in the source method's published configuration, then the magnitude that the intuition refers to has been removed from the mechanism before it can reach the aggregate. Whether anything remains for an allocator to exploit is an empirical and theoretical question that the literature has not settled.

Second, DP and Byzantine robustness are usually studied separately, and are assumed to compose. They interact. DP noise is what a poisoning attacker can hide

behind, and averaging, the operation that a robust aggregator replaces, is what suppresses DP noise. Neither interaction is captured by studying either mechanism alone.

1.3 Problem Statement

Let T communication rounds each consume ρ_t of a client's total budget ρ, with per-round noise multiplier σ_t determined by ρ_t. The allocation problem is to choose $\{\rho_t\}$ with $\sum_t \rho_t = \rho$ so as to maximise final model utility.

The literature treats this as an open optimisation to be solved by a learned policy. We ask instead whether it has a closed-form solution, and if so whether the learned policies can improve on it. We then ask what the same architecture's security properties are under adaptive model poisoning, and whether standard Byzantine defences remain appropriate once DP is present.

1.4 Research Objectives

O1. Characterise the optimum of the per-round privacy budget allocation problem analytically under per-sample clipping, rather than treating it as an open optimisation to be solved numerically.

O2. Test that characterisation empirically in a regime constructed so that accuracy is genuinely sensitive to the noise multiplier, using a statistical protocol precise enough to resolve the effect sizes at stake.

O3. Establish whether learned allocators, as published, can escape the resulting bound.

O4. Measure the honest update envelope of a differentially private detector and determine whether the camouflage premise of DP-exploiting poisoning attacks holds at detector scale.

O5. Evaluate detection-based and aggregation-based Byzantine defences in the presence of DP, including their cost to honest participants.

1.5 Contributions

1. **An optimality result.** Under per-sample clipping, uniform allocation minimises the noise variance accumulated into the global model, and every non-uniform schedule is strictly worse (Section 3.2). The same inequality closes allocation across layers and across clients.

2. **Empirical validation at scale.** Thirteen allocations from three unrelated families, paired with common random numbers, are ordered by a single scalar (Spearman $\rho = -0.767$, $p = 0.0022$). No allocation beats uniform at any privacy budget tested (Section 5.1). The learned allocator itself, trained and evaluated on held-out data over paired trials, likewise fails to beat uniform (Section 5.1.6).

3. **A configuration methodology for DP-FL detection.** We give an SNR model, eff. SNR $\propto \sqrt{B}/(\sigma\sqrt{D})$, that predicts where a detector's accuracy becomes noise-sensitive, and validate its predictions to within a factor of two (Section 4). We also identify a pitfall: adaptive optimisers such as Adam normalise the noise scale away entirely, rendering the privacy parameter inert.

4. **A concentration result for DP poisoning.** High-dimensional concentration makes the honest update norm highly predictable (0.10% relative standard deviation), inverting the premise of DP-camouflage attacks and capping attacker amplitude (Section 5.2.1).

5. **A Byzantine-resilience study under DP**, including the finding that classical robust aggregators are counterproductive in this regime (Section 5.2).

1.6 Scope and Limitations

Results are obtained on a clear-sky subset of the Anti-UAV600 thermal benchmark [12], partitioned by recording session, with a frozen publicly pretrained trunk and a 566,789-parameter privately trained head. Absolute detection numbers describe this subset and should not be read as general anti-UAV performance; the comparisons are relative and unaffected. The poisoning study covers two attacks at up to 30% malicious clients. Section 6.4 states what is and is not covered.

1.7 Thesis Organisation

Chapter 2 reviews differentially private federated learning, adaptive budget allocation, Byzantine-robust aggregation, and poisoning attacks that exploit DP noise, and identifies the gap addressed here. Chapter 3 formalises the system model and proves the optimality of uniform allocation, together with its extensions across layers and clients. Chapter 4 describes the dataset, architecture and calibration procedure used to construct a regime in which the allocation question is measurable at all, and states the statistical protocol. Chapter 5 reports the allocation results and the Byzantine-resilience study. Chapter 6 discusses implications, limitations and future work.

Chapter 2

Literature Review

2.1 Differentially Private Federated Learning

Federated averaging [6] trains a shared model without pooling data, but the updates themselves leak information. DP-SGD [2] closes that gap by clipping per-sample gradients to a norm C and adding Gaussian noise of scale σC, giving a per-step guarantee that composes across steps. In FL this composition runs over communication rounds, so a client with a fixed total budget faces a scheduling decision. Accounting is commonly expressed in zero-concentrated DP [3], under which per-round costs add. All DP mechanisms in this work are implemented with Opacus [10].

The utility cost of DP is severe for dense prediction tasks. Where a classifier must place a single label, a detector must localise objects that may occupy a handful of pixels, and the gradient signal supporting that localisation is small relative to the injected noise. This is the regime in which the present work operates, and Section 4 shows that it is also the regime in which most configurations are uninformative: DP-training an entire detector destroys it, while DP-training only a small head on frozen public features makes the noise irrelevant.

2.2 Adaptive Privacy Budget Allocation

Chen et al. [13] formulate per-round allocation as a decentralised partially observable Markov decision process, with one agent per client, state (t, ρ_k), continuous action $\sigma \in [\sigma_{\min}, \sigma_{\max}]$, and reward the change in global accuracy. They employ DDPG [4] with parameter sharing and a value-decomposition network [7], and report gains over uniform, arithmetic and exponential baselines: 91.05% versus 88.03% on MNIST and 47.75% versus 47.6% on CIFAR-10.

Two features of that setup matter here. The reported CIFAR-10 gain is 0.15 percentage points, roughly 0.3% relative, and is published without error bars; and the agent's state contains no model feedback, so for a fixed budget its policy is a deterministic function of t, that is, a fixed schedule. The optimality result of Section 3.2 therefore bounds the published method directly rather than merely bounding hand-designed schedules.

A related line of work adapts the noise scale from observed training dynamics rather than from a learned policy over the budget. Wang et al. [11] propose DNO-RL, in which a reinforcement-learning agent monitors gradient norms and loss variation and modulates the Gaussian noise scale online, reporting a better privacy-utility trade-off than fixed-rate schedules. Because its state includes model feedback, this family is not strictly covered by our bound; Section 6.5 returns to that distinction, which is the one route by which a learned controller could still escape the result proved here.

2.3 Byzantine-Robust Aggregation

FedAvg has breakdown point zero: a single client can move the mean arbitrarily. Robust aggregators replace the mean with a statistic that tolerates outliers. Coordinate-wise median and trimmed mean [8] operate per-coordinate; Krum [5] selects the single update closest to its nearest neighbours; geometric-median variants interpolate between the two. Their guarantees are derived without differential privacy in the loop, and Section 5.2 shows that this assumption matters: under DP these aggregators discard the averaging that suppresses the injected noise, and can cost more accuracy than the attacks they defend against.

An orthogonal family detects and excludes anomalous updates before averaging. Li et al. [15] evaluate such schemes systematically, clipping update norms adaptively with a threshold set to the median of historical norms and then clustering by cosine similarity; the approach removes the need for manual threshold tuning but degrades sharply under strongly non-IID partitions. Song et al. [16] reduce updates with UMAP before clustering, reporting complete detection of sign-flipping and LIE attacks at substantial server-side cost. Zafar et al. [19] avoid the need for a root dataset by training a conditional GAN on the server to synthesise validation data. Common to this family is that detection quality is reported far more often than the false-positive cost borne by honest clients, a gap the present work addresses explicitly in Table 5.5.

Decentralised variants remove the server entirely. Fang et al. [14] propose a robust local aggregation filter over a peer-to-peer graph that tightens as training proceeds, tolerating a large malicious fraction but assuming benign-subgraph connectivity and providing no protection against leakage during neighbour exchange.

2.4 Poisoning that Exploits DP Noise

Hossain et al. [17] observe that DP's own noise gives an attacker cover: a detector must widen its acceptance band to admit honest noisy updates, and that width is where adversarial perturbation can hide. They introduce α-MPELM, an adaptive attack that tunes a poisoning degree γ from memorised loss history, and show that it deceives norm, accuracy and mixed detectors more effectively than random malicious device (RMD) attacks. Their evaluation uses small models. Section 5.2.1 shows that the premise is dimension-dependent: the relative spread of the honest update norm scales as $1/\sqrt{2D}$, so the hiding room the attack relies on shrinks as the model grows, and at detector scale it is absent.

A parallel concern is the adaptivity of the adversary rather than its concealment. Szeląga et al. [18] show that clients which re-enter the federation while switching attack strategies defeat defences tuned to a single fixed threat, though their evaluation is confined to small feedforward models and IID partitions.

2.5 Federated Reinforcement Learning under Privacy Constraints

Reinforcement learning appears in this literature in a second role, as the learning task being federated rather than the controller of the privacy budget. Zhuo et al. [9] share only noised outputs of local Q-networks, reaching centralised policy quality without exchanging raw observations, but requiring shared transition identifiers and demonstrating only two-party federations. Zhou et al. [21] apply local DP to edge resource allocation and observe that the injected noise can act as a regulariser, while conceding that utility degrades as the number of noisy edge hosts grows. Hafeez et al. [20] combine DP-SGD with homomorphic encryption for robotic-assisted surgery, with fixed privacy parameters and a cooperative threat model. These works establish that DP and federated RL compose in practice, but none of them treats the allocation of the budget over rounds as the object of study.

2.6 Gap Addressed

The allocation literature optimises a quantity whose optimum has not been characterised, and evaluates candidate policies at effect sizes comparable to the run-to-run variance of the experiments used to measure them. The poisoning literature assumes a camouflage effect whose magnitude depends on a model dimension that is not controlled for. This thesis addresses both: it solves the allocation problem in closed form and tests the solution under a paired protocol precise enough to resolve the effect sizes at stake, and it measures the honest update envelope directly at a dimension three orders of magnitude above the models in which the camouflage premise was established.

Chapter 3

Preliminaries, System Model and Theory

3.1 System Model

N clients hold disjoint data shards. In round t the server broadcasts global parameters w_t; each client runs E local epochs of DP-SGD with batch size B and noise multiplier σ_t, and returns its update $\Delta_i^{(t)}$. The server aggregates and forms w_{t+1}.

Within a local epoch, each per-sample gradient g_i is clipped to a norm C,

$$\bar{g}_i \;=\; g_i \cdot \min\!\left(1, \frac{C}{\|g_i\|_2}\right), \tag{3.1}$$

the clipped gradients are averaged over the batch, and Gaussian noise of scale $\sigma_t C$ is added before the local parameter update. Per-round privacy cost in zero-concentrated DP [3] is

$$\rho_t \;=\; \frac{2E}{\sigma_t^2}, \qquad \text{equivalently} \qquad \sigma_t \;=\; \sqrt{\frac{2E}{\rho_t}}, \tag{3.2}$$

and the client's constraint is $\sum_{t=1}^{T} \rho_t = \rho$. Because zCDP costs add across rounds, the allocation $\{\rho_t\}$ is exactly the object a scheduler controls.

3.2 The Optimality of Uniform Allocation

3.2.1 Statement

Theorem 3.1 (Uniform allocation is optimal)**.** *Assume (i) per-sample clipping binds, so each round contributes at most C of signal regardless of σ_t; (ii) the noise injected in distinct rounds is independent; and (iii) utility is non-increasing in the noise variance accumulated into the final model. Then, for fixed T and $\sum_t \rho_t = \rho$, the accumulated noise variance is minimised by $\rho_t = \rho/T$ for all t, and strictly increased by any departure from it.*

Proof. Noise enters the model additively and independently across rounds, so the accumulated variance is proportional to

$$V(\{\rho_t\}) \;=\; \sum_{t=1}^{T} \sigma_t^2 \;=\; 2E \sum_{t=1}^{T} \frac{1}{\rho_t}. \tag{3.3}$$

By assumption (i) the signal term is independent of $\{\rho_t\}$, so the allocation affects utility only through V. Since $x \mapsto 1/x$ is strictly convex on $\mathbb{R}_{>0}$, Jensen's inequality, equivalently the arithmetic-harmonic mean inequality, gives

$$\sum_{t=1}^{T} \frac{1}{\rho_t} \;\geq\; \frac{T^2}{\sum_t \rho_t} \;=\; \frac{T^2}{\rho}, \tag{3.4}$$

with equality if and only if all ρ_t are equal. Hence V attains its unique minimum $2ET^2/\rho$ at the uniform allocation, and any departure strictly increases it. □

The result is deliberately stated in terms of accumulated variance rather than final accuracy, because that is the quantity the mechanism determines exactly. Assumption (iii) is what connects it to utility, and it is the weakest of the three: it is an assumption of monotonicity, not of any particular functional form. Section 5.1.5 tests it directly near the collapse threshold, where it is most vulnerable, and finds that it weakens but never inverts.

3.2.2 Why the Motivating Intuition Fails

The adaptive-allocation argument assumes that the signal contributed per round varies with training progress. Assumption (i) removes exactly that variation: under per-sample clipping every round contributes at most C, whether its true gradients are large or small. The premise is therefore not merely unsupported in this setting, it is incompatible with the DP-SGD mechanism the method is built on.

This is an empirical claim about a measurable quantity, not a modelling convenience, and Section 4.4 measures it in two settings: our own detector, where clipping binds by a factor of 27, and the small convolutional network and clipping norm used by the source method on CIFAR-10, where it binds by a factor of 8. It also shows that the per-sample gradient norm *grows* rather than shrinks over training in that configuration, so the quantity the intuition relies on moves in the opposite direction to the one assumed before clipping discards it.

3.2.3 Extension to Other Allocation Axes

The same inequality closes two further axes along which a budget might be allocated, neither of which requires an additional experiment.

Across layers. Partition the model into groups g of D_g parameters with per-group multipliers σ_g. The zCDP cost of a Gaussian release with per-group scales is $\rho = \frac{1}{2}\sum_g \sigma_g^{-2}$ and the accumulated variance is $V = \sum_g \sigma_g^2 D_g$. Minimising V subject to the budget gives $\rho_g \propto \sqrt{D_g}$ and $V_{\min} = \left(\sum_g \sqrt{D_g}\right)^2 / 2\rho$, whereas a single global clip achieves $D_{\text{tot}}/2\rho$. Since $\left(\sum_g \sqrt{D_g}\right)^2 \geq \sum_g D_g$ by Cauchy–Schwarz, with equality only in the degenerate single-group case, *splitting is always worse*: one pays composition for every group released separately. For the k_1/k_2/head partition used here the best possible per-layer scheme is 2.37× worse than a single global clip (1,345,614 against 566,789 in units of accumulated variance).

Across clients. With FedAvg over N clients the accumulated variance is

$$V \;=\; \frac{1}{N^2}\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_{i,t}^2, \tag{3.5}$$

subject to a *separate* constraint $\sum_t \rho_{i,t} = \rho_i$ for each client, since a privacy budget is a property of a data owner and cannot be transferred between them. The objective therefore separates across i, so each client independently faces Theorem 3.1 and should be uniform in time at its own level. There is no cross-client structure for a credit-assignment mechanism such as a value-decomposition network to discover.

3.2.4 Consequence for Learned Allocators

A learned policy cannot exceed the best allocation available to it. The published agent's state (t, ρ_k) contains no model feedback [13], so for a fixed budget its policy is a deterministic function of t, that is, a fixed schedule, which is exactly the class Theorem 3.1 covers. The bound applies to the method as published, not merely to the hand-designed schedules tested in Chapter 5. Section 5.1.6 confirms this empirically by training the agent and running it against the baselines it was designed to beat.

Reward shaping does not change this. Potential-based shaping provably preserves the optimal policy [1], so it cannot move the optimum; and redefining the reward to something other than utility optimises a different objective, which answers a different question. The only escape is a strictly larger policy class, for example a state containing model feedback such as the current loss or realised gradient norm, which can react to the noise actually drawn rather than committing to a schedule in advance. That class is not covered by the theorem, and Section 6.5 identifies it as the natural next question.

Chapter 4

Experimental Design

This chapter describes the configuration in which the allocation question is measurable at all. That is not a formality: most natural configurations of a differentially private detector are uninformative, either because the noise destroys the model or because it does not reach it, and in both cases every schedule scores alike. Constructing a regime in which accuracy is genuinely sensitive to the noise multiplier is a precondition for the results in Chapter 5, and the procedure for finding it is a contribution in its own right.

4.1 Dataset and Partitioning

We use the Anti-UAV600 thermal infrared benchmark [12]. Sequences are grouped by *recording session*, identified by the timestamp prefix: clips sharing a prefix are the same recording seconds apart, so splitting them across train and test leaks near-duplicate frames. We found this leak in an earlier folder-level split, in which 5 of 11 evaluation sessions also appeared in training, and eliminate it here. A second leak was found and removed in the public pretraining pool, 4 of whose 20 folders shared a session with the evaluation subset; the trunk is therefore re-pretrained on a session-disjoint 97-folder pool.

Experiments use a 29-sequence clear-sky subset spanning 17 sessions. On cluttered sequences the detector saturates near 0.0–0.10 mAP, and degradation cannot be measured against a floor: every schedule scores alike. Removing that floor effect is what makes the allocation question measurable. The final split is $5 \times 3{,}808$ training frames across five clients and 1,500 evaluation frames, disjoint by session.

Restricting to clear-sky sequences raises absolute accuracy and is therefore a favourable choice, but it favours the *alternative* hypothesis rather than ours: it is the condition under which a non-uniform schedule has the most room to demonstrate an advantage. The negative result of Chapter 5 is obtained in the setting most generous to the method it fails to support.

4.2 Model: Public Trunk, Private Head

The architecture follows from a tension established by direct measurement. DP-training an entire detector destroys it, yielding 0.0–0.10 mAP at every usable noise

level, while DP-training only a small 12,677-parameter head on frozen public features makes the noise irrelevant: we measured no effect at all from $\sigma = 0$ to $\sigma = 512$, that is, from $\varepsilon = \infty$ down to $\varepsilon \approx 0.104$. The informative regime lies between these extremes.

We therefore pretrain a convolutional trunk on a session-disjoint public pool, freeze and cache its first three stages s_1–s_3, and privately train the two subsequent stages k_1, k_2 together with the detection head, giving $D = 566{,}789$ differentially private parameters. Note that D is what is *privatised*, not what is randomly initialised; the distinction matters for the initialisation choice below.

Two implementation details proved essential, and both are reported because neither is obvious in advance.

- **SGD, not Adam.** Under Adam a sixteen-fold change in σ produced no change in accuracy (0.336 against 0.320), because per-coordinate normalisation rescales noise-dominated gradients to fixed-size steps. Under SGD the same noise gives monotone degradation (0.311, 0.311, 0.247 at $\sigma = 0.5, 2, 8$). Adaptive optimisers make the privacy parameter inert, which is a silent failure: the privacy accountant still reports the intended ε while the noise never reaches the weights. Figure 4.1 shows both panels.
- **Initialisation.** Training k_1, k_2 and the head from scratch collapses to all-background at every learning rate tried. Seeding k_1, k_2 from the public trunk while leaving the head random keeps optimisation stable and preserves dynamic range, with a floor of 0.001 and a no-DP ceiling of 0.769 $\text{IoU}_{.25}$.

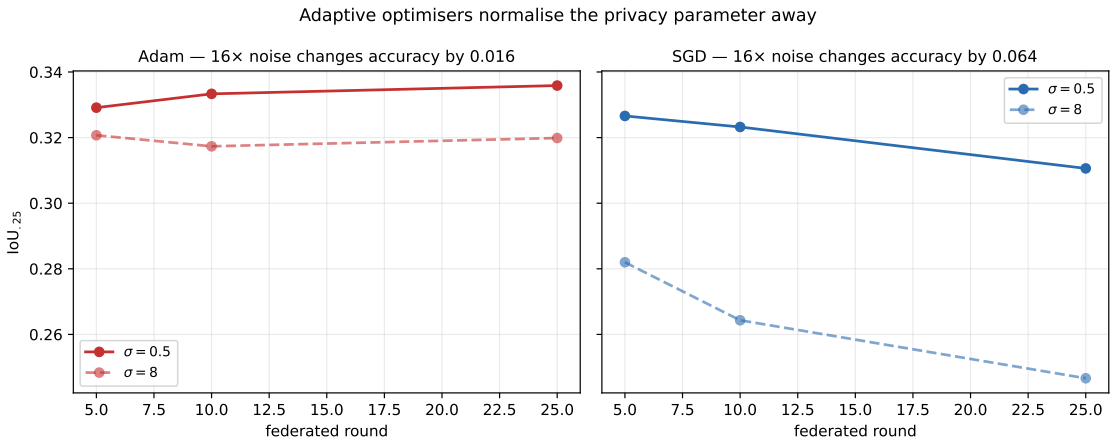


Figure 4.1: Adaptive optimisers normalise the privacy parameter away. A sixteen-fold change in σ moves final accuracy by 0.016 under Adam and 0.064 under SGD, with the SGD degradation monotone in both noise and rounds.

4.3 Locating the Informative Regime

Effective signal-to-noise ratio per round scales as

$$\text{eff. SNR} \;\propto\; \frac{B}{\sigma\sqrt{D}}\sqrt{\text{steps} \cdot N} \;\propto\; \frac{\sqrt{B}}{\sigma\sqrt{D}}, \tag{4.1}$$

since the number of steps scales as $1/B$. Signal accumulates coherently across steps and clients while noise accumulates as a square root, which is why the averaging gain enters as $\sqrt{\text{steps} \cdot N}$.

Calibrating the constant on the head-only configuration, where the knee sits at $\sigma \approx 512$ for $D = 12{,}677$ and $B = 2048$, predicts a degradation knee at $\sigma_{\text{knee}} \approx 1.69\sqrt{B}$ for $D = 566{,}789$. The predictions of 38 at $B = 512$ and 19 at $B = 128$ were borne out by measured onsets of 32–64 and 16–32 respectively, so the model is accurate to within a factor of two and is usable for choosing B in advance rather than by sweeping. We adopt $B = 128$, SGD with learning rate 0.003 and momentum 0.9, $C = 1$, and $T = 25$ rounds of $E = 2$ local epochs.

Table 4.1 gives the resulting operating curve, and Figure 4.2 plots it against the source method's operating range. A trivial constant-output predictor scores 0.021, confirming that the metric is not degenerate; targets are well spread, with the most common cell occupied in 4.6% of frames.

Table 4.1: Noise sensitivity in the calibrated regime. $\text{IoU}_{.25}$ on held-out sessions; floor (random head) 0.001, trivial predictor 0.021.

σ	0	2	4	8	16	32
$\text{IoU}_{.25}$	0.769	0.557	0.499	0.492	0.371	0.047

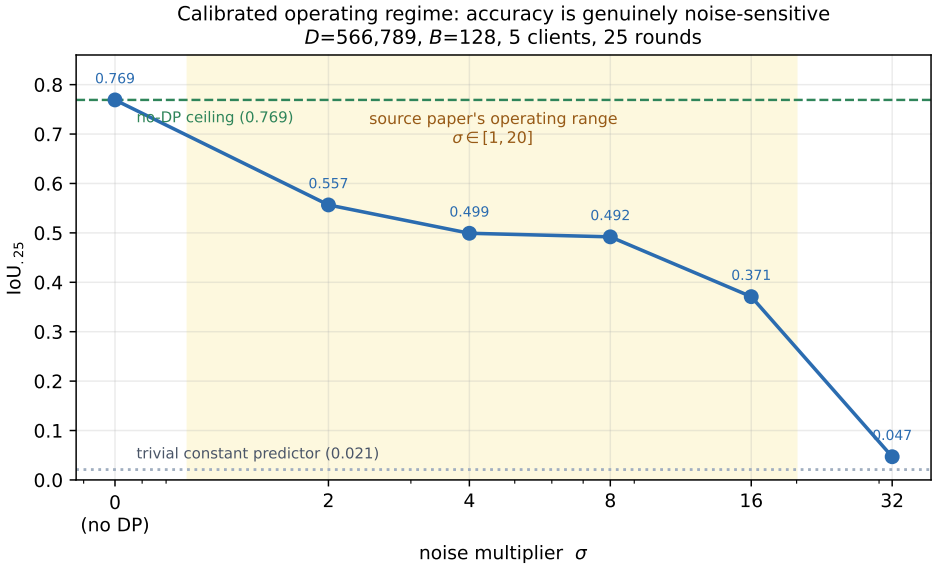


Figure 4.2: The calibrated operating regime. Accuracy is genuinely sensitive to the noise multiplier across the source method's operating range $\sigma \in [1, 20]$, which is the precondition for any allocation study. Floor and ceiling are marked.

4.4 Verifying the Theorem's Assumption

Theorem 3.1 rests on assumption (i): that per-sample clipping binds, so each round's signal contribution is pinned at C. This is not an idealisation to be taken on trust.

It is measurable, and it is the single quantity that determines whether adaptive allocation has anything to work with. Table 4.2 reports median per-sample gradient norms against the clipping threshold, for our detector and for the small convolutional network and $C = 6$ used by the source method on CIFAR-10.

Table 4.2: Does clipping bind? Median per-sample gradient norm against the clipping threshold C. Clipping binds in both settings, so assumption (i) of Theorem 3.1 holds for the source method's own configuration as well as for ours.

Setting	Stage	Median $\|g_i\|$	C	Binds by
CIFAR-10 CNN ($D = 59{,}386$)	initialisation	1.18	6.0	—
	after 5 rounds	37.93	6.0	6.3×
	after 25 rounds	48.11	6.0	8.0×
Anti-UAV head ($D = 566{,}789$)	steady state	26.85	1.0	26.8×

Two observations follow. First, clipping binds in *both* settings, by 8× on CIFAR-10 and 27× on the detector, so the difference between them is one of degree rather than of regime, and the optimality result applies to both. At initialisation the CIFAR-10 gradients sit below the threshold and no clipping occurs, but the fraction of clipped samples rises to 0.90 within five rounds and remains high thereafter, so the binding regime is reached almost immediately and holds for all but the opening rounds.

Second, and more directly damaging to the motivating intuition, the per-sample gradient norm *grows* over training on CIFAR-10, from 1.18 at initialisation to 48.11 after 25 rounds. The adaptive-allocation argument assumes the opposite, that gradients are large early and shrink as the model converges, so that noise should be front-loaded. Under cross-entropy on a small network the per-sample gradient norm increases with training as the model grows confident and the loss surface sharpens, and in any case clipping removes the variation before it can reach the aggregate. The premise is therefore doubly unavailable: the quantity it relies on moves the wrong way, and is then discarded by the mechanism.

4.5 Threat Model for the Poisoning Study

The aggregation server is honest but curious: it follows the protocol and does not inject updates of its own, but may inspect what it receives, which is the motivation for client-side DP. Up to 30% of clients are malicious and may submit arbitrary updates. Malicious clients know the global model, the detection rule in force and the DP parameters, which is the strongest information assumption under which a norm-preserving attack is meaningful. They cannot observe honest clients' private data or the server's held-out validation set. Section 5.2.1 derives what this threat model permits an attacker to do, and the answer constrains the attacks worth implementing.

4.6 Statistical Protocol

Comparisons are *paired*: all schedules within a trial share one initialisation and one random-number stream, a common-random-numbers (CRN) design, so the noise draws are correlated across arms and initialisation luck cancels. CRN reduced the standard deviation of paired differences from 0.0466 to 0.0171, a 3.2× tightening, so that 20 paired trials carry the precision of roughly 150 without it. Scores are the mean $\mathrm{IoU}_{.25}$ over the final rounds, rounds 20–25, rather than a single round, since single-round scores oscillate near the collapse threshold.

Budget fairness is checked explicitly rather than assumed. Effective ρ spent is 30.0 for every arm except $k = -1$, which underspends at 26.1 and is therefore conservative, and $k = +3$ at 29.6. The $\sigma \leq 20$ clip touches only rounds carrying negligible allocations, so no arm buys an advantage by saturating it.

This protocol matters more than it might appear. Four separate single-run results in this project did not survive replication: an apparent +0.0107 for $k = +1$, +0.0023 for $k = -1$, a +0.230 excursion at a tight budget, and the original flat noise curve that was later traced to a head-initialisation artefact. Cliff-adjacent single runs carry essentially no information, and we report effect sizes with confidence intervals throughout rather than point estimates alone.

4.7 Implementation Provenance

The source method was ported from the authors' description and run on CIFAR-10 before being transferred to detection, as a check that the implementation behaves sensibly on the data it was designed for. Across the four allocation arms the final accuracies obtained were 0.4364, 0.4383, 0.4251 and 0.4063, in the range the original work reports, but each arm is a single run and the spread between arms is comparable to the run-to-run variance documented in Section 4.6. We therefore do *not* present this as a replication, and no claim in this thesis rests on it: at $n = 1$ per arm the comparison cannot distinguish a real difference from noise in either direction, which is the same criticism we apply to the published result.

The decisive check that our conclusions are not an artefact of a faulty port is instead the direct measurement in Section 4.4. That measurement is made on the source method's own architecture, clipping norm and dataset, and it establishes the premise of Theorem 3.1 in that configuration independently of any accuracy comparison.

Chapter 5

Results

5.1 Allocation Across Rounds

All results in this section use the calibrated regime of Chapter 4 and the paired protocol of Section 4.6. The quantity Theorem 3.1 identifies as governing the outcome, the accumulated noise variance $\sum_t \sigma_t^2$, is reported alongside every schedule as a multiple of the uniform minimum, so that the theory's prediction can be read directly against the measurement.

5.1.1 Power-Law Schedules

We allocate $\rho_t \propto (t+1)^k$, normalised to spend $\rho = 30$ exactly. Here $k = 0$ is uniform, and $k > 0$ places less noise late, which is the direction predicted by the adaptive-allocation literature. Table 5.1 reports $n = 20$ paired trials.

Table 5.1: Power-law schedules against uniform, $n = 20$ paired trials with common random numbers, $\rho = 30$. The variance penalty is $\sum_t \sigma_t^2$ relative to the uniform minimum. Every significant entry is a loss.

k	Variance penalty	Mean difference	95% CI	p
-1.0	$1.99\times$	$+0.0023$	$[-0.0012, +0.0058]$	0.19
$+0.5$	$1.18\times$	-0.0018	$[-0.0041, +0.0006]$	0.13
$+1.0$	$1.98\times$	-0.0045	$[-0.0125, +0.0035]$	0.25
$+2.0$	$10.15\times$	-0.0214	$[-0.0447, +0.0018]$	0.069
$+3.0$	$21.03\times$	-0.0539	$[-0.0878, -0.0201]$	**0.0034**

The ordering is monotone in k, and every step toward the literature's prescription is worse, significantly so at the extreme. Only $k = -1$ sits above uniform, and it is not significant; Section 5.1.2 shows that it does not replicate.

5.1.2 A Mild Front-Load

Because $k = -1$ and $k = +1$ carry near-identical variance penalties ($1.99\times$ against $1.98\times$) yet differed in sign, the possibility remained of a genuine timing effect distinct from the variance term. If such an effect existed, it could be bought most cheaply at

small negative k, where the variance penalty is negligible ($k = -0.25$ costs $1.04\times$). We pre-registered that test and ran it on disjoint seeds.

At $n = 16$ paired trials the effect is absent: $k = -0.25$ gives -0.0007 with 95% CI $[-0.0018, +0.0003]$, and $k = -1$ does not replicate, moving from $+0.0023$ to -0.0013. **Any gain from mild front-loading is bounded above by** $+0.0003$ $\mathbf{IoU}_{.25}$, equivalently 0.04% relative, which is the tightest null in this study. The apparent asymmetry on which the hypothesis rested was itself noise.

5.1.3 Beyond the Power Family

A power law is a narrow family, and an optimality claim should not rest on it. Bang-bang schedules, consisting of a run of minimum-noise rounds with the remaining budget spread over the rest, in both orderings, and random draws from the budget simplex also lose to uniform, by -0.047 to -0.112. Early stopping, spending the whole budget over $T' < T$ rounds and halting, is neutral at best: the $\sqrt{T}$ averaging gain exactly offsets the $\sigma \propto \sqrt{T}$ budget cost, measured at $+0.0011$ for $T' = 20$ and -0.058 for $T' = 8$.

5.1.4 One Scalar Explains Everything

Figure 5.1 plots all 13 allocations against their accumulated noise variance. A single scalar orders outcomes across three unrelated families: Spearman $\rho = -0.767$ ($p = 0.0022$), and Pearson on log variance -0.851 ($p = 0.00023$). Uniform sits at the minimum, $1.00\times$, and nothing beats it.

This is the central empirical result. It is stronger than the individual comparisons because it is a prediction about the *ordering* of unrelated schedule families from a single closed-form quantity, and that prediction holds across families that share no structure other than their variance.

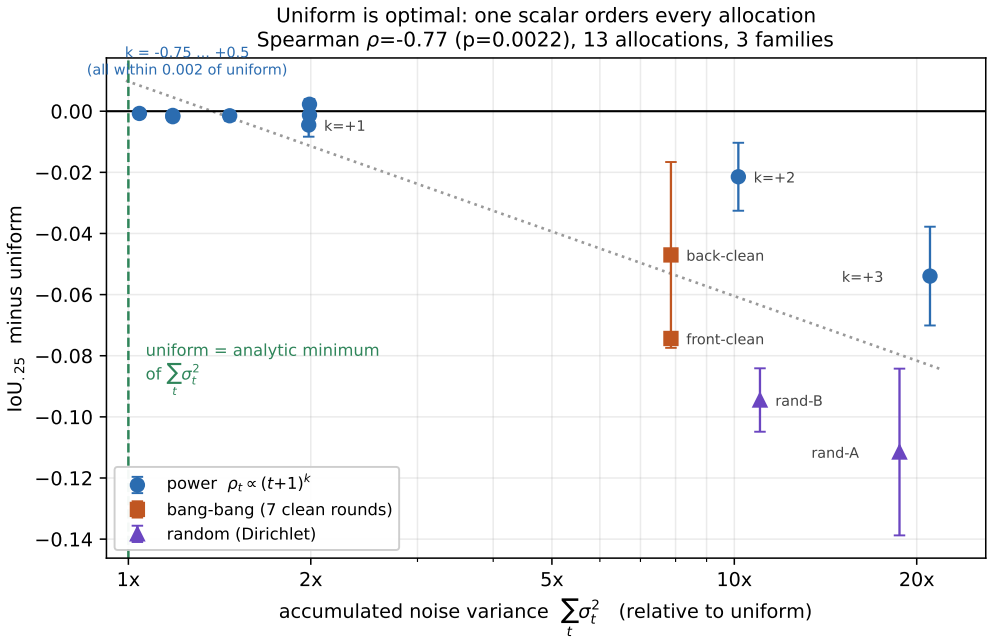


Figure 5.1: Accumulated noise variance $\sum_t \sigma_t^2$ orders every allocation tested, across power-law, bang-bang and random-simplex families. Uniform is the analytic minimum.

5.1.5 Tight Privacy Budgets

Theorem 3.1 assumes utility is smooth and monotone in accumulated variance, an assumption weakest near the collapse threshold, where outcomes become bimodal: a run either survives the noise or fails. If any allocation is to beat uniform, this is where it should happen, because concentrating a tiny budget into a few usable rounds might rescue a run that spreading it thin would lose. We therefore ran a pre-specified ladder at $\rho \in \{0.5, 2, 8\}$ against truncation depths $T' \in \{3, 6, 12, 25\}$, $n = 10$ paired trials, reporting all outcomes (Table 5.2).

Table 5.2: Truncation against uniform across privacy budgets, $n = 10$ paired trials. Entries are the mean IoU$_{.25}$ difference, with wins out of 10 in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$.

ρ (uniform σ)	$T' = 12$	$T' = 6$	$T' = 3$
0.5 ($\sigma = 14.1$)	+0.036 (5/10)	+0.040 (4/10)	−0.050 (4/10)
2.0 ($\sigma = 7.07$)	−0.025 (3/10)	−0.034 (3/10)	−0.123* (2/10)
8.0 ($\sigma = 3.54$)	−0.022 (2/10)	−0.039* (1/10)	−0.128** (1/10)

No allocation significantly beats uniform at any budget, and every significant entry is a loss. At $\rho = 0.5$ the point estimates turn positive, but the win counts of 4/10 and 5/10 are coin flips: the positive mean comes from a few large asymmetric outliers, occasions on which concentrating the budget rescued a run that uniform lost, rather than from consistent superiority.

What does change with budget is the size of the penalty for deviating, which shrinks

monotonically as the budget tightens, with a slope against $\log \rho$ of -0.021 to -0.029 across the three truncation depths. The theorem's smoothness assumption therefore does measurably weaken toward the cliff, exactly as anticipated, but never enough to reverse the ordering. Reporting that gradient inside a null result is the honest characterisation: uniform's margin erodes at tight budgets and is never overturned.

Crucially, the paired standard deviation grows faster than the effect, from 0.060 at $\rho = 8$ to 0.103 at $\rho = 2$ and 0.173 at $\rho = 0.5$, so the best candidate gain of +0.040 sits at less than a quarter of the noise in the very signal an agent would have to learn from. Resolving it would require $n = 73$, roughly 28 hours of computation, and with 4/10 wins that is very likely chasing noise. **Even were the gain real, it would be unlearnable.** Figure 5.2 shows both panels.

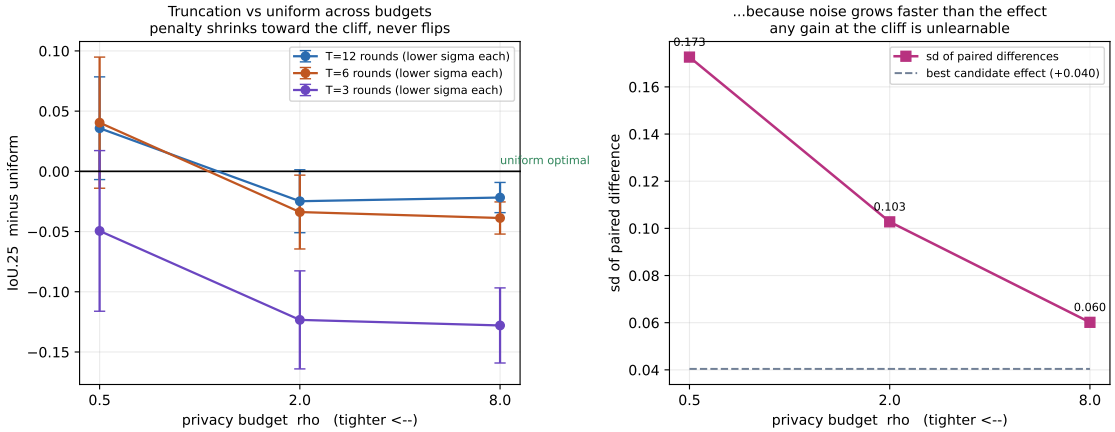


Figure 5.2: Left: the penalty for deviating from uniform shrinks toward the collapse threshold but never flips sign. Right: paired-difference noise grows faster than the candidate effect.

5.1.6 The Learned Allocator in Practice

Sections 5.1–5.1.5 bound the class of fixed schedules, and Section 3.2.4 argues that the published agent belongs to that class because its state carries no model feedback. That argument is structural. It is worth also running the agent, and we do.

We trained the DDPG agent with parameter sharing and value decomposition, as described by Chen et al. [13], and compared it against the uniform, exponential and arithmetic baselines used in that work. Comparisons are paired, with all four arms sharing one initialisation per trial, over $n = 8$ trials, and every point is scored on a held-out validation set that no client trains on. Table 5.3 reports final accuracy and Figure 5.3 the trajectories.

Table 5.3: The learned allocator against the baselines it was designed to beat. Final mAP@0.5 on held-out data, $n = 8$ paired trials, mean $\pm$ standard deviation; differences are DRL minus baseline.

Method	Final mAP@0.5	Difference	95% CI	Wins
DRL (learned)	0.8629 ± 0.0606	—	—	—
Uniform	0.9060 ± 0.0483	-0.0431	$[-0.0997, +0.0136]$	3/8
Exponential	0.8738 ± 0.0666	-0.0109	$[-0.0674, +0.0456]$	4/8
Arithmetic	0.8931 ± 0.0761	-0.0302	$[-0.0761, +0.0157]$	3/8

The agent does not beat any of the three baselines. Every point estimate is negative, and it loses to uniform by -0.0431 with 3 wins in 8 trials, the largest deficit in the table. None of the differences reaches significance at $n = 8$, so the honest statement is that the agent fails to demonstrate an advantage rather than that it is proven inferior; but the direction is consistent across all three comparisons and consistent with Theorem 3.1.

Figure 5.4 shows why. The agent learns a genuinely non-uniform schedule, which is what it was designed to do, and by Theorem 3.1 that is precisely what costs it accuracy: every departure from the flat allocation raises $\sum_t \sigma_t^2$ without buying any additional signal, since clipping has already pinned the per-round contribution at C. The agent is not failing to learn; it is successfully learning to do the thing that does not help.

These runs use the lightweight convolutional pipeline of an earlier stage of this project rather than the 566,789-parameter two-stage detector used elsewhere in this chapter, so the absolute accuracies are not comparable to Table 4.1 and the two sets of numbers should not be read against each other. The comparison within the table is unaffected: all four arms share an architecture, an initialisation and a data split.

An evaluation artefact worth recording. An earlier version of this comparison showed the DRL agent beating uniform in 5 trials out of 5. That result was scored on each client's *own training shard*. Re-scored on held-out data, the same policies lost, 0.823 against 0.906. The figures and table above use the held-out numbers throughout. We record the discrepancy because it is a fifth instance, alongside the four replication failures noted in Section 4.6, of a favourable result in this problem that did not survive a correction to the measurement, and because scoring on data a client has trained on is an easy error to make when each client holds a different shard.

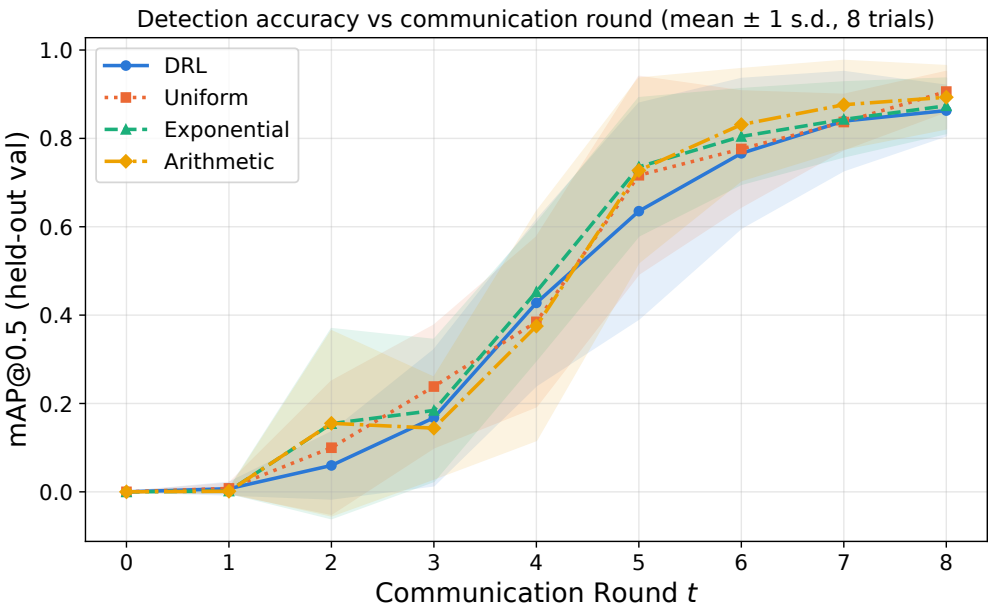


Figure 5.3: Detection accuracy against communication round for the learned allocator and the three fixed baselines, mean over 8 paired trials with ±1 standard deviation bands, scored on held-out data.

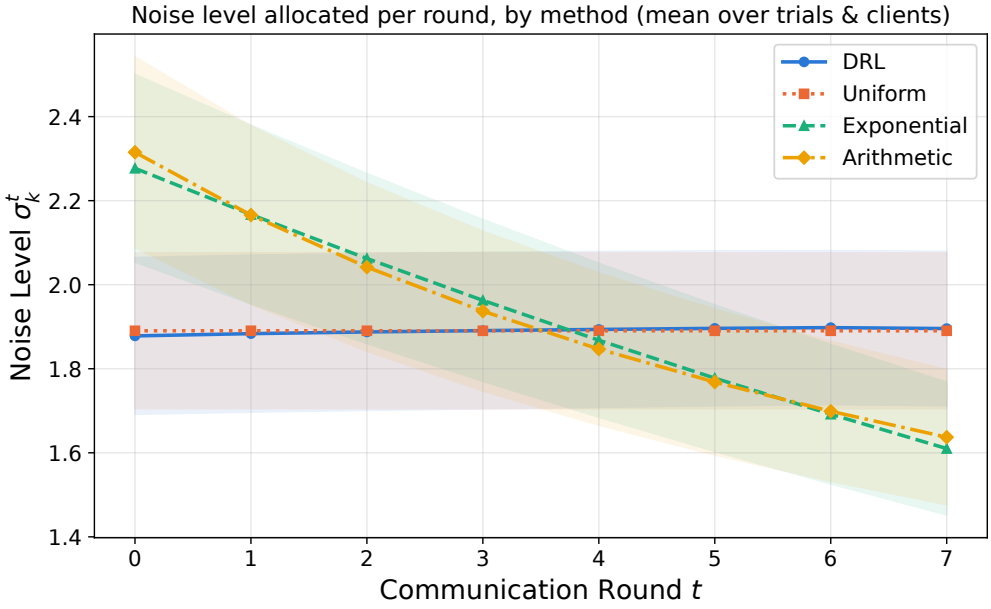


Figure 5.4: The noise schedule the agent actually learned, against the three fixed schedules. The learned policy is non-uniform, which by Theorem 3.1 is the reason it underperforms.

5.1.7 A Note on What Is Not Being Claimed

The single largest effect measured anywhere in this project was a +0.230 advantage, +46% relative, for $T' = 6$ at $\rho = 0.5$ in the first trial of the ladder. It did not survive: $n = 2$ gave +0.071, $n = 6$ gave −0.0003, and $n = 10$ gave +0.040 with 4/10

wins. We report it because it illustrates the failure mode this chapter is designed to exclude, and because the same phenomenon plausibly explains small positive results elsewhere in the literature.

5.2 Byzantine Resilience under Differential Privacy

5.2.1 The Honest Update Envelope

Every defence needs a model of what a legitimate update looks like, because that is what an attacker must imitate, and it is what sets the amplitude an attacker can use. Table 5.4 measures it directly.

Table 5.4: Honest client update norms. Relative spread collapses once DP noise is present, matching the $1/\sqrt{2D} = 0.094\%$ concentration of a Gaussian norm at $D = 566{,}789$.

σ	Update norm	Relative s.d.
0 (no DP)	0.6917 ± 0.6548	95%
1.83	2.2487 ± 0.0187	0.83%
4.0	4.8853 ± 0.0095	0.19%
8.0	9.7586 ± 0.0100	0.10%

This inverts the premise of DP-exploiting poisoning. Attacks of that family assume DP noise widens the acceptance band an attacker can hide in. In high dimensions the norm of a Gaussian vector concentrates, so DP noise makes the update norm *more* predictable, not less: without noise the norm varies by 95% across clients, and with it by 0.1%, against a theoretical prediction of 0.094%. The camouflage effect is a small-D phenomenon and vanishes at detector scale.

What survives is *direction*. A norm detector constrains magnitude only, and in 566,789 dimensions a rotated update is nearly orthogonal to the honest one while carrying an indistinguishable norm. We therefore implement the adaptive attack as a norm-preserving direction attack, which is the strongest form the envelope permits.

5.2.2 Attacks and Defences

RMD returns a randomly falsified update with no attempt at concealment. **α-MPELM** steers against the honest update and rescales to the honest norm, adapting its poisoning degree γ according to whether it was detected in the previous round.

Defences are a **norm** detector using a median-absolute-deviation band; an **accuracy** detector using server-side validation influence on a held-out set of 300 frames, disjoint from the 1,200-frame set on which reported metrics are computed; their **mix**; and three robust aggregators, **Krum** [5], **coordinate-wise median** and **trimmed mean** [8]. The grid comprises 33 configurations $\times$ 3 trials, with 10 clients of 1,904

frames each, $m \in \{1, 2, 3\}$ malicious clients, that is 10 to 30%, at $\sigma = 1.83$ corresponding to uniform allocation at $\rho = 30$, over 15 rounds.

5.2.3 Results

Table 5.5: Byzantine resilience at $m = 30\%$ malicious clients. $\text{IoU}_{.25}$ over final rounds; TPR is the detection rate against α-MPELM; FPR is the fraction of honest clients wrongly rejected.

Defence	No attack	RMD	α-MPELM	TPR	FPR
None	0.6485	0.6410	**0.6312**	—	0.00
Norm	0.6521	0.6446	0.6428	0.14	0.10
Accuracy	0.6656	0.6635	0.6616	0.24	0.19
Mix	0.6655	0.6583	**0.6649**	**0.32**	0.28
Krum	0.6386	**0.5759**	0.6760	0.00	0.00
Median	0.4402	0.4333	**0.3168**	0.00	0.00
Trimmed mean	0.6441	0.6373	0.6282	0.00	0.00

Three findings follow, illustrated in Figures 5.5–5.8.

The published attack asymmetry reproduces. The norm detector catches RMD at a true-positive rate of 1.00 and α-MPELM at 0.14. The conventional attack is caught every time and the adaptive one walks through, which is the headline result of the source work, reproduced here on a real detector. The episodic adaptation behaves as designed: γ escalates to 1.0 when the attacker is never caught, and is driven down to approximately 0.14 once a detector begins firing (Figure 5.7).

The architecture is inherently resilient. The worst undefended degradation is -0.017 $\text{IoU}_{.25}$, or 2.7% relative, at 30% malicious clients. This follows from the envelope result: with a roughly 2% norm band an attacker cannot amplify, so the strongest legal move is direction reversal at matched norm, and three reversed updates in ten merely scale the average down rather than corrupting its direction. Resilience here is a *consequence of DP*, not of any defence. Mix detection removes the residue entirely, scoring 0.6649 against a 0.6485 no-attack baseline, at the cost of rejecting 28% of honest contributions. That trade should be reported alongside detection rate, since a defence that discards a quarter of legitimate participants is not free.

Classical robust aggregators are counterproductive under DP. Coordinate-wise median costs -0.208 $\text{IoU}_{.25}$ *with no attacker present*, an order of magnitude more than any attack studied here, and Krum costs -0.073 under RMD. The reason is structural: averaging is precisely what suppresses DP noise, by a factor of $\sqrt{N}$, and median and Krum discard that averaging while also breaking the update's directional structure. Under DP, the standard Byzantine toolkit can cost more than the threat it addresses. Krum under α-MPELM scores 0.6760, the highest cell in the

table, because selecting the single most central update happens to exclude direction-reversed outliers, but the same mechanism loses badly to RMD, so it is not a reliable defence.

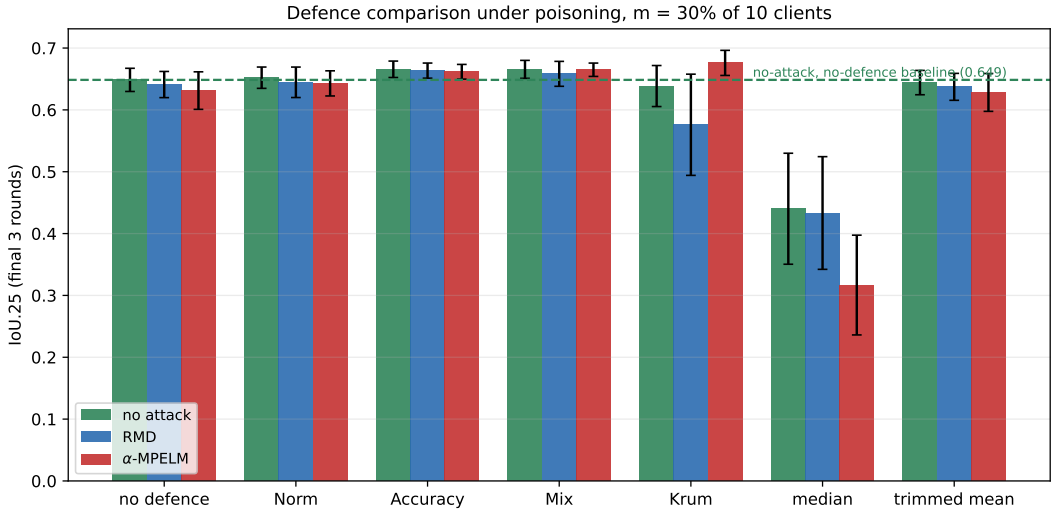


Figure 5.5: Defence comparison at $m = 30\%$ malicious clients.

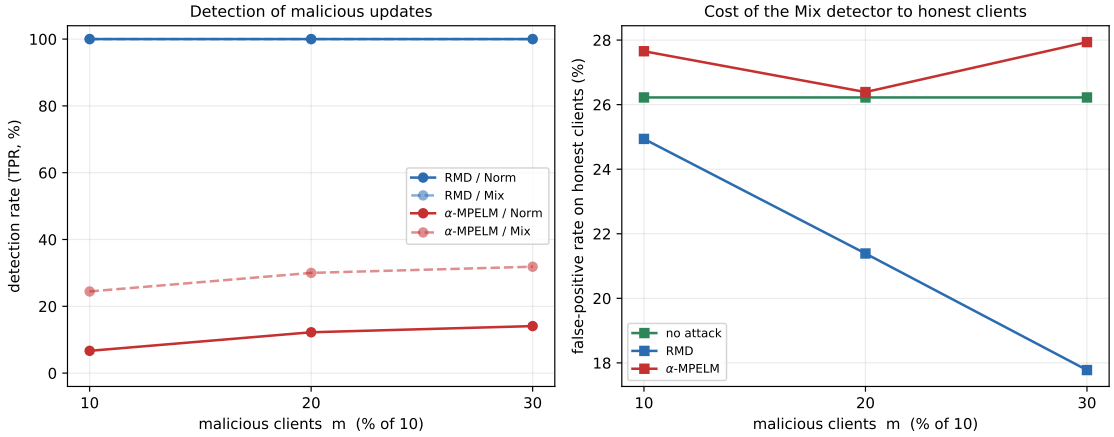


Figure 5.6: Detection rate against malicious ratio, and the false-positive cost borne by honest clients.

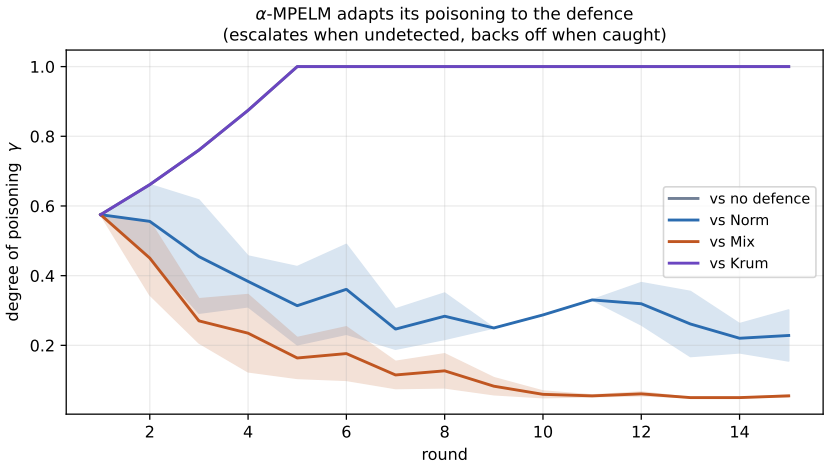


Figure 5.7: Poisoning degree γ over rounds: the attacker escalates when undetected and retreats when caught.

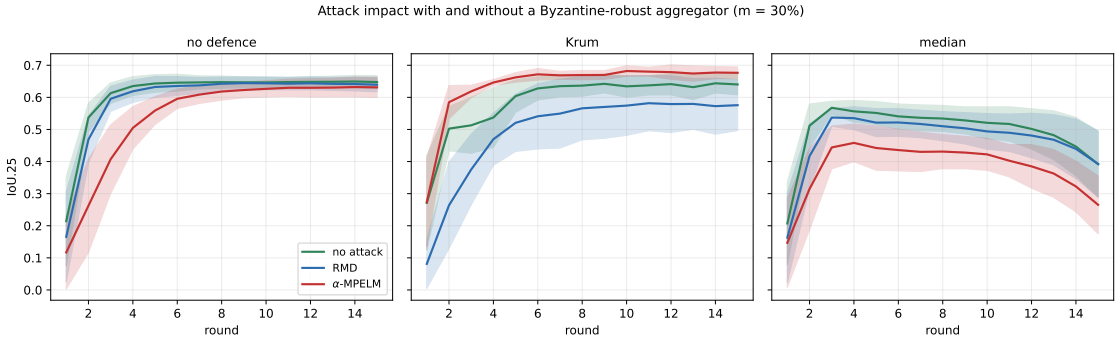


Figure 5.8: Accuracy trajectories with and without robust aggregation.

Chapter 6

Discussion and Conclusion

6.1 Summary of Findings

Adaptive per-round privacy budget allocation cannot outperform uniform allocation in clipped DP-FL, because clipping removes the very signal variation the approach is designed to exploit. We prove this (Theorem 3.1), verify it across 13 allocations from three unrelated families ordered by a single closed-form scalar, and show that the bound applies to published learned allocators because their state space confines them to fixed schedules. Training the agent and running it confirms the prediction: it learns a non-uniform schedule and loses to uniform by -0.043 mAP over paired held-out trials. The same inequality closes the analogous problems of allocating a budget across layers and across clients, neither of which requires a separate experiment.

On the security side, the same high dimensionality that makes DP costly for detection also makes it protective. Update norms concentrate as $1/\sqrt{2D}$, attacker amplitude is capped by the resulting narrow envelope, and the observed degradation under adaptive poisoning at 30% malicious clients is 2.7% relative. Detection-based defences neutralise what remains; classical robust aggregators cost more than the attacks do.

6.2 Implications for Practice

Three recommendations follow for practitioners building differentially private federated systems.

1. **Spend the privacy budget uniformly** and invest engineering effort elsewhere. Uniform is not merely a reasonable default but the analytic optimum under per-sample clipping, and the effort required to search for a better schedule is better spent on the architecture or the data partition.

2. **Do not use adaptive optimisers with DP-SGD.** Adam normalises the privacy parameter away: a sixteen-fold change in σ moved accuracy by 0.016, while the privacy accountant continued to report the intended ε. This is a silent failure that inflates apparent utility at a claimed privacy level.

3. **Prefer detection over robust aggregation when DP is present**, and report the false-positive cost. Averaging is what suppresses DP noise, so

replacing it with a robust statistic forfeits that suppression; in our setting coordinate-wise median cost an order of magnitude more than any attack tested.

6.3 Implications for the Literature

The results suggest that reported gains from adaptive allocation, often around 0.3% relative and published without error bars, are within the noise of the experimental designs used to obtain them. The CRN protocol used here reaches at $n = 20$ the precision that $n \approx 150$ would give without it, and at that precision the effect is absent.

We are careful about what this does and does not say. The claim is not that the original work is mistaken about what it observed. The claim is that the observation is not separable from noise at the precision used to obtain it, and that a mechanism argument, Theorem 3.1 together with the measurement in Table 4.2 showing that clipping binds by 8× even in the source configuration, predicts that no separable effect should exist.

A negative result of this kind is only as strong as the effort spent trying to make the positive result appear. That effort is documented rather than asserted: Chapter 5 reports a pre-registered search over three schedule families, a mild front-load chosen specifically to be cheap in variance, early stopping, and a ladder of privacy budgets down to the collapse threshold, with all outcomes reported including those that initially looked favourable. Four separate single-run results that appeared positive did not survive replication, and each is reported.

6.4 Limitations

Results are obtained on a clear-sky subset partitioned by recording session, and absolute numbers do not transfer to cluttered scenes, where the detector sits near the floor. We note that this choice favours the alternative hypothesis: it is the condition under which a non-uniform schedule has the most room to show an advantage.

Theorem 3.1 assumes that clipping binds, verified here at 27× for the detector and 8× for the source configuration, and that utility is monotone in accumulated variance. The latter assumption measurably weakens near the collapse threshold, where the penalty for deviating from uniform shrinks with slope -0.021 to -0.029 against $\log \rho$, though never enough to change the conclusion. Configurations in which clipping does not bind, for instance a very large C with small gradients, fall outside the theorem; there the premise would be live, though such configurations also forfeit much of the privacy guarantee that motivates clipping.

The poisoning study covers two attacks at $m \leq 30\%$ malicious clients, at a single noise level, on the clear-sky subset. The cross-client extension of the theorem is proved but not verified empirically. The implementation port to CIFAR-10 described in Section 4.7 is single-run per arm and no conclusion rests on it.

6.5 Future Work

The one policy class our bound does not cover is an agent whose state includes model feedback, such as the current loss or realised gradient norm, and which can therefore react to the noise actually drawn instead of committing to a schedule in advance. Theorem 3.1 constrains the expected accumulated variance of any fixed schedule, but a controller that observes the realised trajectory is optimising a different, larger object. Whether exploitable per-run variance exists at all is unmeasured, and is the natural next question; the online noise-modulation family exemplified by DNO-RL [11] is the closest existing point of comparison.

On the security side, an attacker that optimises direction against the accuracy detector, rather than simply reversing it, would be a stronger test than the one used here, since the accuracy detector is the component that carries the mix defence. Extending the envelope analysis across model scales would also establish where the camouflage premise crosses from live to inert, which this work locates only at its two endpoints.

6.6 Conclusion

The contribution is a negative result with a proof and a positive result with a mechanism. Uniform privacy-budget allocation is optimal, not merely a reasonable default, and the learned allocators proposed to improve on it are bounded by that optimum because their state space confines them to the class of fixed schedules. The same dimensional scaling that makes differential privacy expensive for detection also makes the resulting system difficult to poison, by concentrating the honest update norm and capping the amplitude available to an attacker. Both findings were obtained in a regime constructed specifically to give the alternatives every chance to succeed.

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